

# CS1540 Algorithmic design

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Homework 3.

## 1 Introduction

These exercises are simply for additional practice and include an implementation task that can be automatically tested via CodeGrade.

For testing via Codegrade see the link [Canvas](#). Your program needs to read a filename from the terminal, read the input from the named file, and print the output (and nothing else) to the terminal. The Canvas page contains an example `.java` file demonstrating the I/O. Note that **the file you upload must have the same name and main class as the example, otherwise the testing will fail**. An example input file and its expected output is also included. You can submit as many times as you want. To ensure your implementation is fast, the runtime time limit for each test case is 20 seconds.

## 2 Weg afgesloten!

You are travelling in a region which consists of  $T$  towns, which are connected by a set of  $R$  roads; each road  $r$  joins two towns, and (depending on distance, road surface quality, etc.) takes some amount of time  $d_r$  to travel along. You want to travel from town  $a$  to town  $b$ , setting out at some time  $t_{\text{dep}}$ . We want to find the earliest time  $t_{\text{arr}}$  at which you can arrive at  $b$ .

Unfortunately, due to high rainfall the roads in this region require constant maintenance, resulting in frequent road closures— $C$  of them have been planned during your visit. You will need to take account of these road closures when planning your route.

The first line of the input file will consist of integers  $T R C a b t_{\text{dep}}$ , with the following restrictions:

- $T, R, C, a, b$ , and  $t_{\text{dep}}$  of size at least 1
- $T, R, C, a, b$ , and  $t_{\text{dep}}$  of size at most  $10^6$
- $1 \leq a \leq T$
- $1 \leq b \leq T$
- $a \neq b$

- There may be multiple roads between the same pair of towns
- A road may have multiple closures. For example, given  $R_{12} = (58, 162, 7)$ , there may be two closures  $C_1 = (12, 11, 15)$  and  $C_2 = (12, 47, 58)$ .
- Road closures may overlap. For example, given  $R_{12} = (58, 162, 7)$ , there may be two closures  $C_1 = (12, 35, 50)$  and  $C_2 = (12, 47, 58)$ .

The next  $R$  lines will describe the roads: each will consist of three integers, the start and endpoints of the road and the time required to travel along it. Note that roads can be used in either direction. The roads are numbered 1 to  $R$  in the order they are given.

The following  $C$  lines will describe the road closures. Each will consist of three integers: the number of the road being closed, the start time of the road closure and the end time of the road closure.

Your program should output a single integer, the earliest possible arrival time  $t_{\text{arr}}$  at town  $b$ . If it is impossible to travel from town  $a$  to town  $b$ , you should output 0.

Note that road closures apply to the whole of a road, so you must have left the road by the time a closure begins. On the other hand, if a closure ends at time  $t$ , you can start travelling on the road at time  $t$ . So for example, suppose a particular road takes time 4 units to travel along, and has a closure from time 6 to time 8. Then we could start travelling along this road at time 2, or time 8, but not at times 3,4,5,6 or 7. Of course, you are allowed to stop for coffee in towns en route (for instance if you need to wait for a road to open).

Name your program **Closed.java** for submission to Codegrade.

For example, if the region has just two towns connected by a single road which takes time 4 and has two closures (3,6) and (5,8), and we want to travel from town 1 to town 2 leaving at time 1, the input file would be

```
2 1 2 1 2 1
1 2 4
1 3 6
1 5 8
```

and the expected output is 12.

### 3 Your tasks

1. Write an algorithm (in pseudocode) to solve the problem above efficiently. Justify the correctness and analyse the running time of your algorithm.
2. Implement your algorithm in Java and submit it to Codegrade.
3. Formulate a decision problem version of this problem and write down the formal language it corresponds to.
4. Is this language in NP? Is it in P? Justify your answers.

5. Suppose now that each road closure has a price  $p_i$  for which you can bribe the local gemeente to cancel it. Given that you have a total budget  $B$  for paying bribes, you again want to find the earliest possible arrival time. Show that this problem is NP-complete. (Hint: to show that it is NP-hard, you may want to use a reduction from the Knapsack problem)